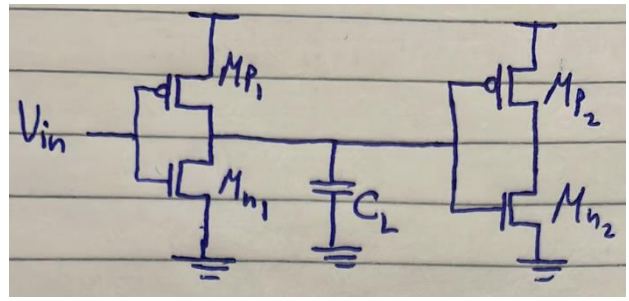
What we studied so far is the static characteristics of the CMOS. That is, we applied an input and waited for a long time until everything settled (i.e. all caps became open circuit). Now, we study the dynamic characteristics of the CMOS.

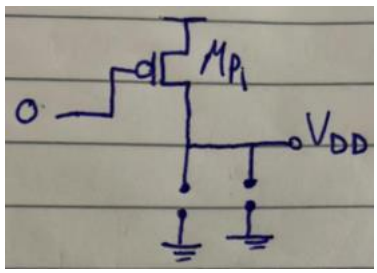
Delay

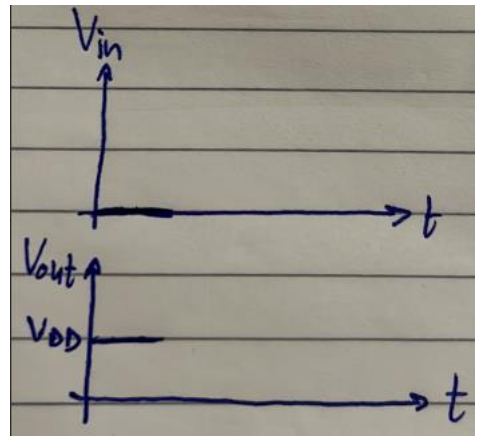
Assume we have a driver inverter and a load inverter as shown on the right. We are concerned with the first inverter only and want to calculate its delay, the second inverter is just a load. We know that there is a capacitance at the output node of the first inverter (we will determine it exactly soon). This capacitance is from:



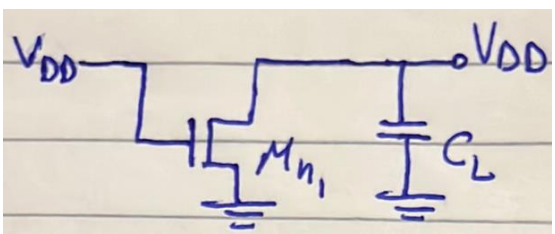
1) Inverter 1 drain nodes.
2) Inverter 2 gate nodes.

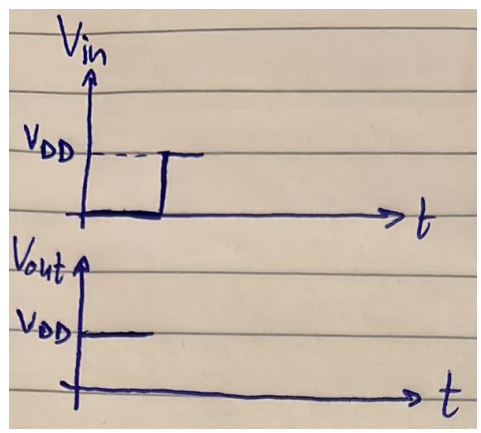
Consider the case that V_{in} was logic 0 V for a long time. Thus, the equivalent circuit is:





Now, assume the input has abruptly risen to V_{DD}. At this instant, the equivalent circuit is:





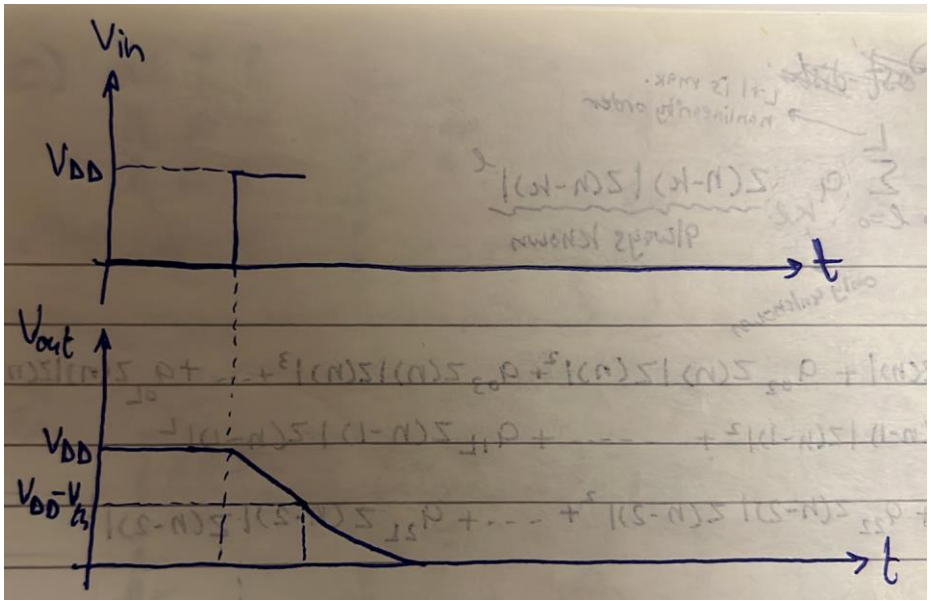
How will the output behave?

M_{n1} is in saturation region, so it conducts current $I = \frac{K}{2}(V_{DD} - V_{TH})^2$. Hence, the capacitor will discharge the charge on it ($Q = C_L V_{DD}$) through the NMOS and hence the output voltage will begin decreasing. But how will it decrease (i.e. linearly or quadratically)? Writing KCL at the output:

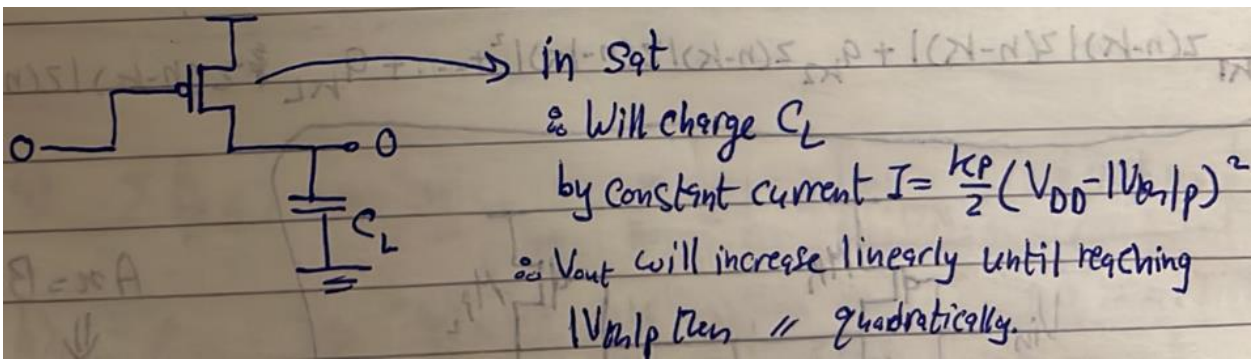
$$I_{C_L} = I_{NMOS}$$

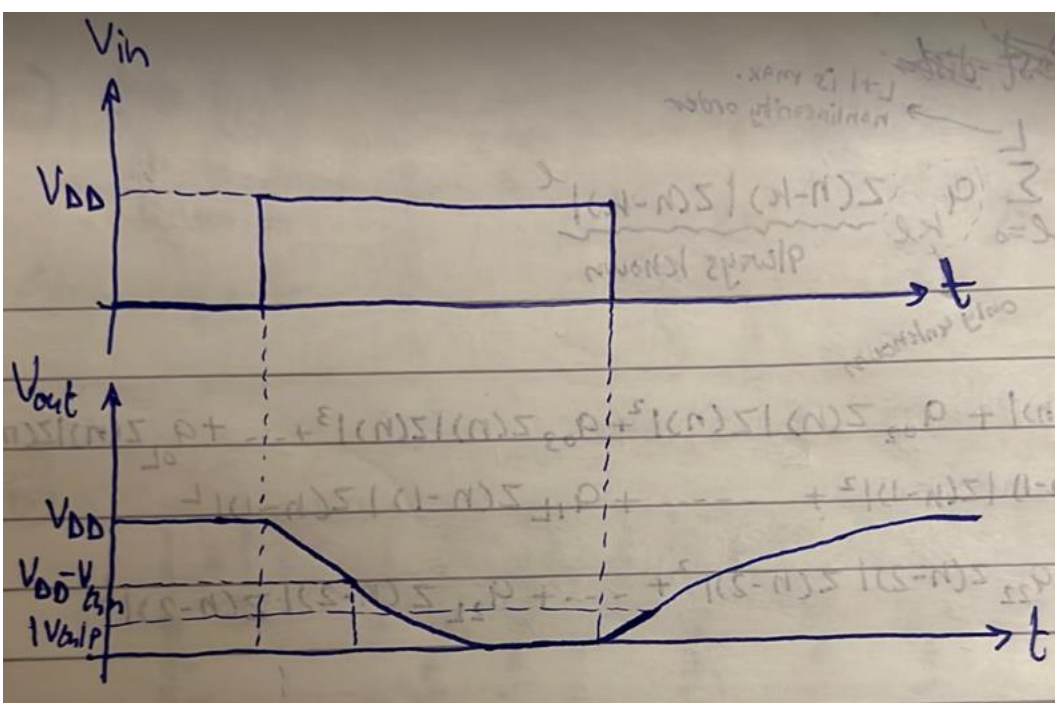
$$C_L \frac{dV_{out}}{dt} = \frac{K_n}{2}(V_{DD} - V_{THn})^2 \rightarrow \frac{dV_{out}}{dt} = \frac{\frac{K_n}{2}(V_{DD} - V_{THn})^2}{C_L} = constant$$

Since the slope is constant, V_{out} will begin decreasing linearly. But when V_{out} reaches $V_{DD} - V_{TH}$, the NMOS will become in triode and hence I_{NMOS} is no longer constant (since it changes with V_{DS}) and hence V_{out} will begin decreasing quadratically:



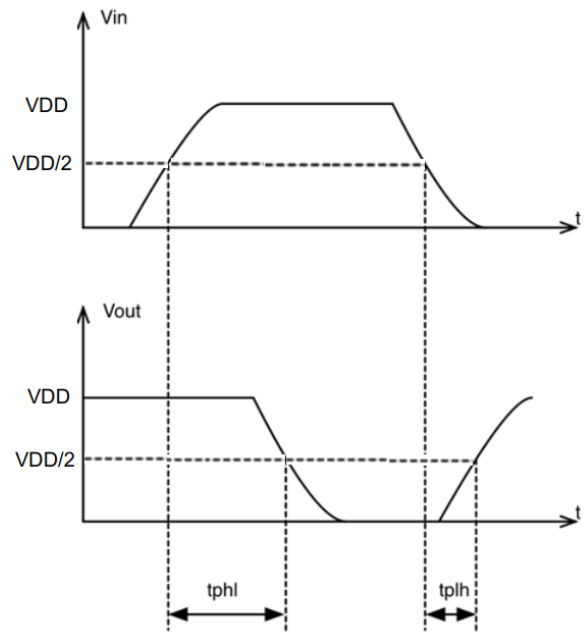
At steady state, the output settles at 0 and the capacitor is open circuit again. Now assume input abruptly changed to 0, the output and the equivalent circuit will be:



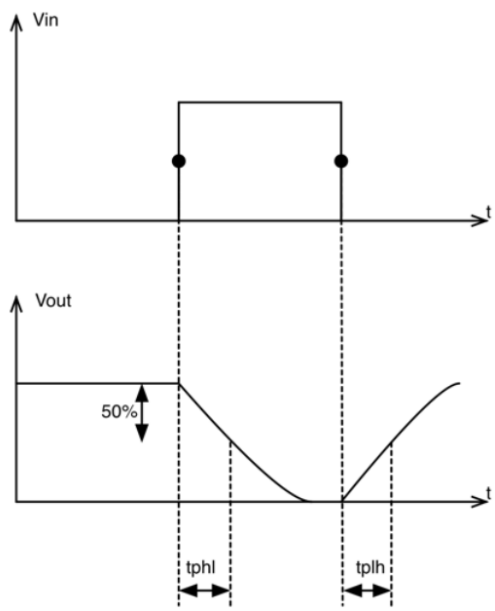


So where is the delay here?

The delay, known as ***propagation delay*** of the logic gate, is defined as the time from the input crossing 50% to the output crossing 50%. There are *high-to-low propagation delay* (t_{phl}), which is when the output changes from '1' to '0', and *low-high propagation delay* (t_{plh}), which is when the output changes from '0' to '1', as shown below:



So, notice that the propagation delay is measured using both the input and output waveforms. However, for simplicity, we will assume that the input changes abruptly as we showed in the previous example. Accordingly, the delay will be as shown below; instead of using both input and output waveforms to determine the propagation delay, it becomes possible to determine it just from the output waveform as the time taken from the instance the output changed (whether from 0% or 100%) to 50%.



Are t_{phl} and t_{plh} the same?

Although the same capacitor C_L charges and discharges, however it charges through PMOS and discharges through NMOS, which have different resistances due to the difference in mobility between NMOS and PMOS. So, how will we calculate t_{phl} and t_{plh}? By using a simple model called "RC Model" which models the transistors as resistors, so that we can draw the two cases as:

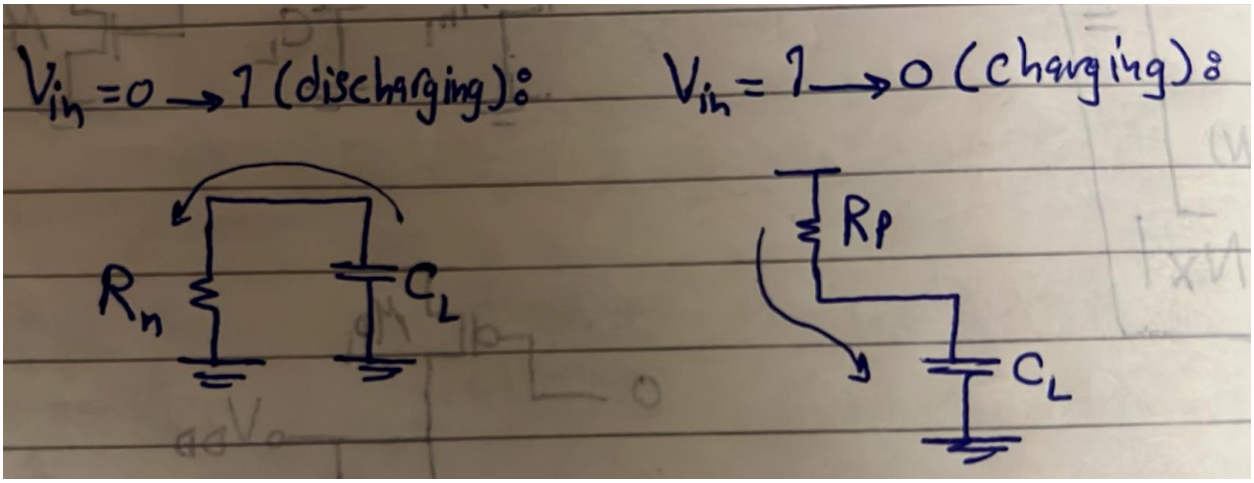
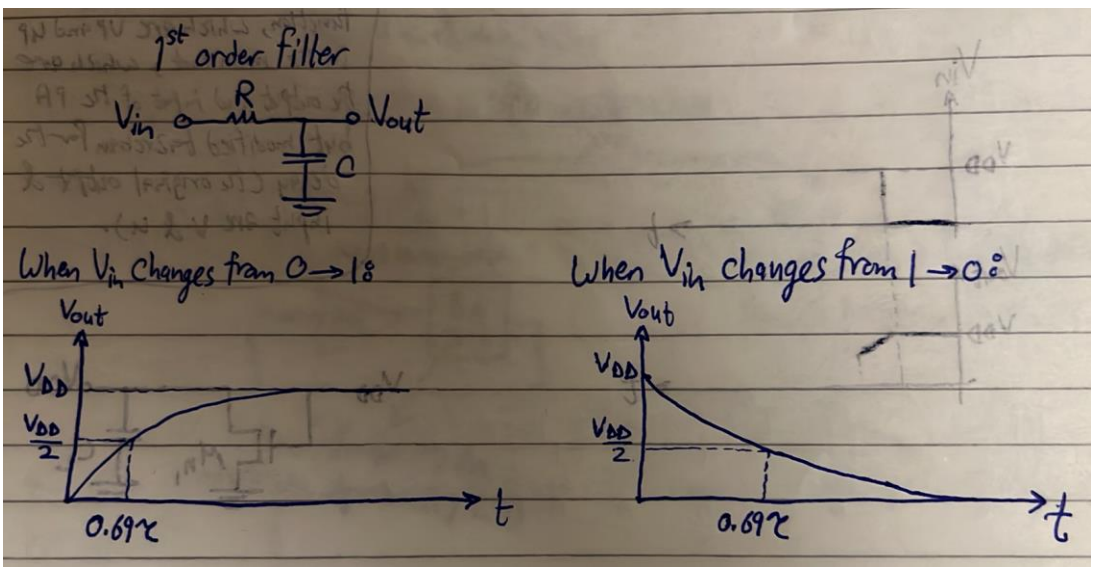


Figure 1

Tamam e3tabarna el transistors ka resistors, bardo hane7seb el delay ezay? Modeling the transistors as resistors 5alla el circuit ba2et just a simple 1st order RC filter which we already know how to calculate its delay. Remember:



Lama el input beta3 el 1st order filter beyetghayar abruptly (i.e. step input) sawa2 men 0 --> 1 or 1 --> 0, el output beya5od wa2t $\ln(2) * \tau = 0.69\,\tau = 0.69RC$ 3ashan yewsal le 50% men el total change. E7na me3arafeen el propagation delay ezay? En howa el wa2t men awel ma el output yetghayar l7d ma yewsal le 50% men el total change. Fa keda ne2dar n2ool en el propagation delay beta3 el 1st order filter da be $0.69RC$.

Tab ma el 1st order filter da howa el case beta3etna bzbt! 3andena resistor, el howa el transistor, in series m3ah capacitor, el howa C_L, w el output meta5ed benhom, w fag2a benghayar el input el da5el 3al resistor da, kol el far2 bs en fel RC filter da benghayar el input f3ln, laken fel inverter el by7sl en ya2ema el pull-up network beteshtaghal fag2a fa yeb2a el da5el ka input howa el V_{DD} wel resistor howa R_p, aw el pull-down network teshtaghal fag2a fa yeb2a el da5el ka input howa le ground wel resistor howa R_n, exactly zy ma7na rasmeen fi figure 1. Thus, for our inverter:

$$t_{phl} = 0.69R_{n1}C_L$$

$$t_{plh} = 0.69R_{p1}C_L$$

So, we deduced an expression for the propagation delay of a CMOS inverter, all what we need now is to know how to calculate the effective resistance of a transistor (to know R_{n1} and R_{p1}) and how to calculate the capacitance C_L and the output of the first inverter.

Capacitance

$$C_L = C_{gn2} + C_{gp2} + C_{dn1} + C_{dp1}$$

$$C_g = C_{ox} WL$$

$$C_d = C_j \, WL_{drain}$$

, where C_{ox}: gate cap/unit area

C_j: drain cap/unit area

طول الـ drain الـ مرسوم
في الـ layout

C_{ox}, C_j, L_{drain} are defined by the technology

Resistance

MOSFET resistance is non-linear as it changes during cutoff, ohmic and saturation regions. So, taking the average and doing some analysis, we derive the following expression:

$$R_{eq} = \frac{3}{4} * \frac{V_{DD}}{I_{D,SAT}}$$

Thus, we obtained an expression for the propagation delay of an inverter.

What is the effect of increasing W and L on the delay?

$-$	C_g	C_d	R_{eq}
$L \uparrow$	$\uparrow$	$-$	$\uparrow$
$W \uparrow$	$\uparrow$	$\uparrow$	$\downarrow$

It is clear that there is no point in increasing L as both resistance and capacitance increase. Thus, we always use $\boldsymbol{L_{min}}$ in digital circuits to minimize delay. Regarding W, it can be noticed that capacitance increases while resistance decreases, so the effect of W is still controversial and needs a discussion. So, from now on, we always refer to the increase in the aspect ratio or the size of a transistor as an increase in its width W since L is constant = L_{min}.

Unit Transistor

A unit transistor is a transistor whose aspect ratio (i.e. $\frac{W}{L}$) equals 1.

- For NMOS:

$$C_{gn_o} = C_{ox}L_{min}^2$$

$$C_{dn_o} = C_j L_{min} L_{drain}$$

$$R_{n_o} = \frac{3}{4}\frac{V_{DD}}{\frac{1}{2}\mu_n C_{ox}(VDD - V_{thn})^2}$$

- For PMOS:

$$C_{gp_o} = C_{ox}L_{min}^2$$

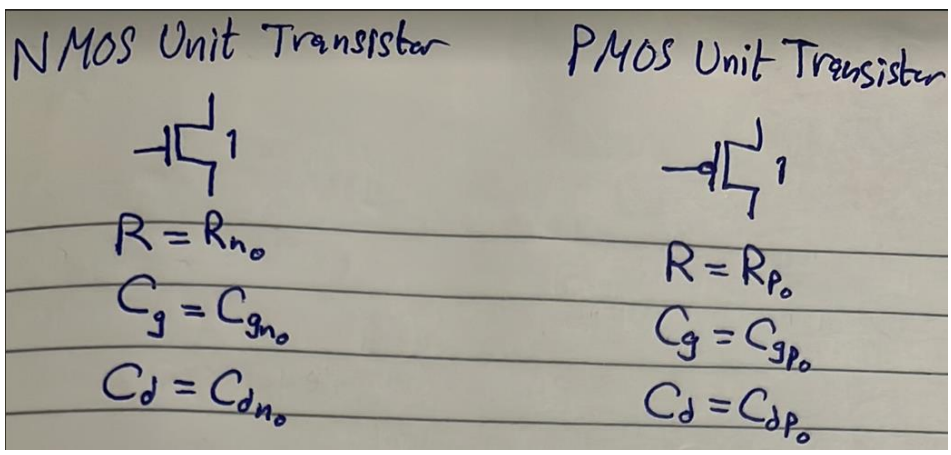
$$C_{dp_o} = C_j L_{min} L_{drain}$$

$$R_{p_o} = \frac{3}{4}\frac{V_{DD}}{\frac{1}{2}\mu_p C_{ox}\left(VDD - |V_{thp}|\right)^2}$$

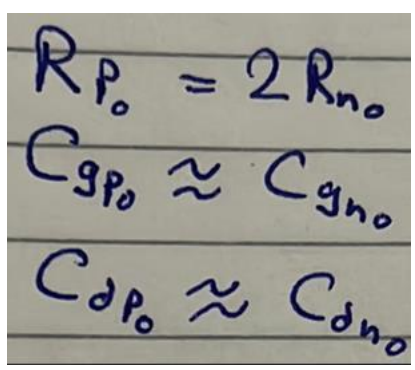
$$assuming\ \mu_p = \frac{\mu_n}{2}\ \&\ |V_{thp}| = V_{thn} \rightarrow R_{p_o} = \frac{3}{4}\frac{V_{DD}}{\frac{1}{2}\frac{\mu_n}{2}\ C_{ox}(VDD - V_{thn})^2} = 2R_{n_o}$$

Notice that R_o, C_{g_o}, C_{d_o} are all technology-dependent and hence we assume they are all given.

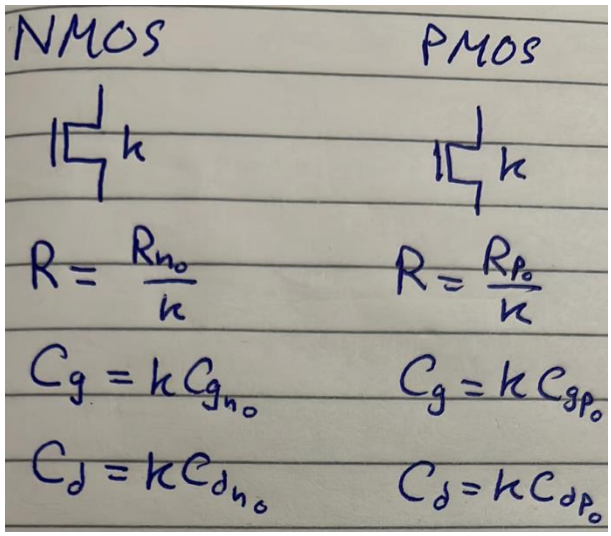
So, we can notice that both NMOS unit transistor and PMOS unit transistor have the same capacitance, the only difference is that the PMOS unit transistor's resistance is double that of the NMOS. However, fel 7al hanfara2 been el NMOS wel PMOS parameters as follows:



W demneyan neb2a 3arfeen:

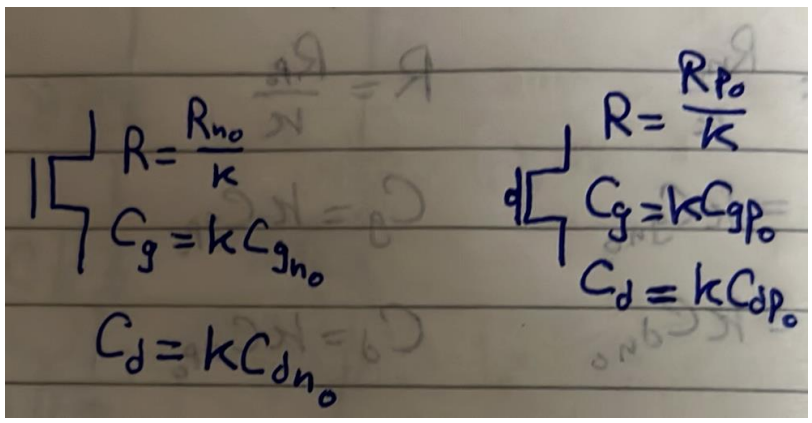


For transistors with any arbitrary size (i.e. aspect ratio) k:

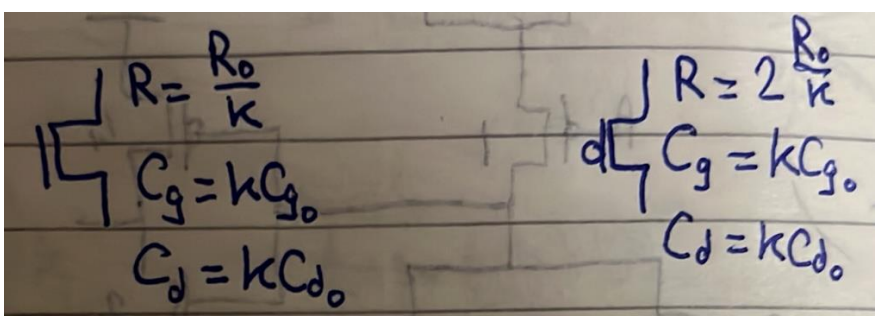


This tells us that, for any transistor, ***its capacitance is directly proportional with its size, while its resistance is inversely proportional with its size (i.e. The larger the transistor, the lower its resistance and the higher its capacitance).***

So, what to always remember howa el rasma de:



W lw 2alak Assume $C_{gn_o} = C_{gp_o} = C_{g_o}$, $C_{dn_o} = C_{dp_o} = C_{d_o}$, $R_{p_o} = 2R_{n_o} = 2R_o$, yeb2a:



For an inverter, what if we want t_{phl} and t_{plh} to be equal?

To have both propagation delays equal, we need both NMOS and PMOS to have the same resistance. Thus:

$$\frac{R_o}{k_n} = 2\frac{R_o}{k_p} \rightarrow k_p = 2k_n$$

For a PMOS to have the same resistance as NMOS, its size should be **double**

Revisiting time-constant equation of the inverter:

$$\tau = R C_L = R(C_{dn_1} + C_{dp_1} + C_{gn_2} + C_{gp_2})$$

hl or lh n_1 or p_1

$$= \underbrace{R(C_{dn_1} + C_{dp_1})}_{\tau_{intrinsic}} + \underbrace{R(C_{gn_2} + C_{gp_2})}_{\tau_{extrinsic}}$$

$\tau_{int.}$: delay due to self-loading

$\tau_{ext.}$: delay due to external loading

This tells us that even if the gate is not loaded, it still has a delay due to its self-loading (called internal/intrinsic/parasitic/unloaded delay).

What happens to the delay of the inverter when its size is increased by ratio k (i.e. when the aspect ratio of the transistors are increased by ratio k?

When the transistor's size increases by ratio k, its resistance decreases by k while its capacitance increases by k. Accordingly:

$$\tau = \frac{R}{\cancel{k}}(\cancel{k} C_{dn_1} + \cancel{k} C_{dp_1}) + \frac{R}{k}(C_{gn_2} + C_{gp_2})$$

$$= R(C_{dn_1} + C_{dp_1}) + \frac{R}{k}(C_{gn_2} + C_{gp_2})$$

Yeb2a lama azawed el size beta3 gate, el resistance beta3t el transistors el fiha bet2el while el capacitance betzeed, fa el intrinsic delay beta3 el gate mesh beyetghayar laken el extrinsic delay bey2el, and hence el overall delay beta3 el gate bey2el. Fa keda ana f3ln 2alelt el delay beta3 el gate beta3ty 3an taree2 eny zawedt el size beta3ha. However, zeyadet el size de zawedet el capacitance beta3ty el heya to3tabar load capacitance 3al gate el ableya, w bel taly keda el delay beta3 el gate el ableya zaad. This tells us that, to minimize the overall delay, we have to think of sizing the whole chain, not a transistor on its own!

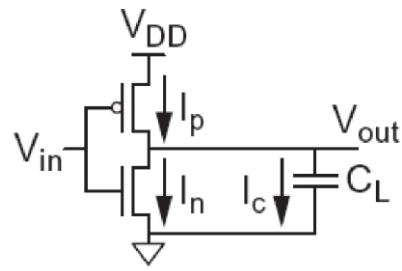
When a gate size is increased, its delay decreases. However, the load on the gate before increases and hence its delay increases.

Also, this result tells us that ***when a gate size is scaled, its intrinsic delay does not change!*** That is, after sizing a gate, multiply this size by any scaling factor, the intrinsic delay will stay the same.

Switching Power

We knew that CMOS does not have static power since its pull-up and pull-down networks are never on at the same time. What about dynamic power (i.e. power consumed during transitioning from '1' to '0' or vice versa)?

When the gate output rises:



- Energy stored in the capacitor is:

$$E_c = \int P\,dt = \int VI\,dt = \int V_{out} C_L \frac{dV_{out}}{dt}\,dt = C_L \int_0^{V_{DD}} V_{out}\,dV_{out}$$
$$= \frac{C_L V_{DD}^2}{2}$$

- But energy drawn from the supply is:

$$E_{V_{DD}} = \int P\,dt = \int VI\,dt = \int V_{DD} C_L \frac{dV_{out}}{dt}\,dt = V_{DD} C_L \int_0^{V_{DD}} dV_{out}$$
$$= C_L V_{DD}^2$$

So the energy drawn from the supply is $C_L V_{DD}^2$, while the energy stored in the cap is $\frac{C_L V_{DD}^2}{2}$. This shows that, during switching from low to high , $\frac{C_L V_{DD}^2}{2}$ of energy were consumed in the PMOS as heat.

When the gate output falls:

- Energy stored in the capacitor (i.e. $\frac{C_L V_{DD}^2}{2}$) is dumped to ground through NMOS and hence $\frac{C_L V_{DD}^2}{2}$ of energy is consumed in the NMOS as heat.

So, when the output made 2 transitions, called 1 switching (i.e. 1 switching = 2 transitions), we drew $C_L V_{DD}^2$ energy from the supply, half of them were dissipated as heat in the PMOS and the other half was dissipated as heat in the NMOS.

- Energy consumed in one switching cycle:

$$E_{sw} = C_L V_{DD}^2$$

- Average switching power (a.k.a dynamic power):

$$P_{sw} = \frac{E_{sw}}{T_{sw}} = C_L V_{DD}^2 f_{sw}$$

Watch this from 29:24 to 34:07.

$$\boldsymbol{P_{sw} = \alpha C_L V_{DD}^2 f}$$

What is the dependence of the delay on the supply V_{DD}?

$$delay \propto RC$$

$$R \propto \frac{1}{I} \propto \frac{1}{V_{DD}}$$

$$\therefore delay \propto \frac{1}{V_{DD}}$$

$$V_{DD} \uparrow \quad delay \downarrow$$

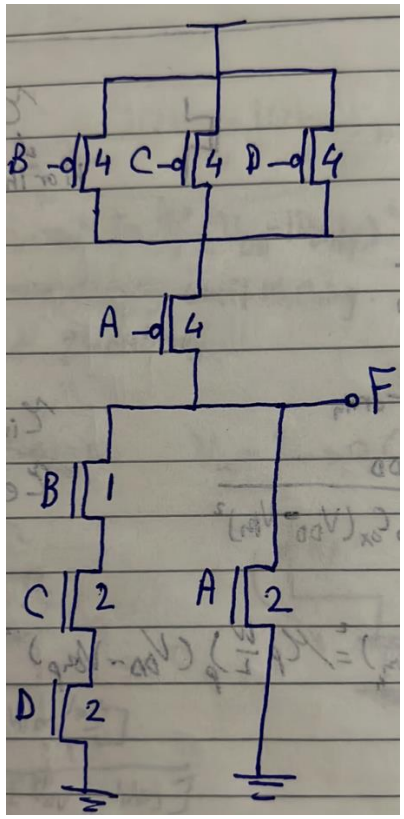
What is the dependence of dynamic power on the supply V_{DD}?

$$P_{dyn.} \propto V_{DD}^2$$

$$V_{DD} \uparrow \quad P_{dyn.} \uparrow$$

Consider the shown circuit. What is the delay when the inputs ABCD change:

1) From 0110 to 1011.
2) From 0110 to 0111.
3) From 0110 to 1111.
4) From 1111 to 0011.



Answer:

The capacitance at the output is:

$$C_L = C_d|_{p,A} + C_d|_{n,A} + C_d|_{n,B} = 4C_{dp_o} + 2C_{dn_o} + 1C_{dn_o}$$

We made an assumption here that we neglected the internal nodes' capacitances, we considered only the caps at the output node. Is that a good approximation? yes, because, as we will see, kol ma bab3ed 3an el output node, ta2seer el caps 3leha bey2el.

1) When ABCD = 0110, the output is high. When the inputs change to 1011, the output will switch to low through NMOS A. Thus, the resistance it will see is:

$$R|_{N,A} = \frac{R_{n_o}}{2}$$

Thus, the high-to-low propagation delay in this case is:

$$t_{phl} = 0.69 * RC_L = 0.69 * \frac{R_{n_o}}{2} * (4C_{dp_o} + 2C_{dn_o} + 1C_{dn_o})$$

2) The output will switch to low through NMOS B, C, D. Thus:

$$R = \frac{R_{n_o}}{1} + \frac{R_{n_o}}{2} + \frac{R_{n_o}}{2} = 2R_{n_o}$$

Thus:

$$t_{phl} = 0.69 * RC_L = 0.69 * 2R_{n_o} * (4C_{dp_o} + 2C_{dn_o} + 1C_{dn_o})$$

3)

$$R = (\frac{R_{n_o}}{2} || 2R_{n_o}) = \frac{R_{n_o}}{2.5}$$

$$t_{phl} = 0.69 * RC_L = 0.69 * \frac{R_{n_o}}{2.5} * (4C_{dp_o} + 2C_{dn_o} + 1C_{dn_o})$$

4) When ABCD = 1111, the output is low. When the inputs change to 0011, the output transitions to high through PMOS B, A:

$$R = \frac{R_{p_o}}{4} + \frac{R_{p_o}}{4} = 0.5R_{p_o}$$

What are the inputs that cause worst-case rise and fall resistances?

1) Rise Resistance: worst-case rise resistance occurs when the output node charges through PMOS A and only 1 PMOS out of the three B, C, D:

$$R|_{rise,worst} = \frac{R_{p_o}}{4} + \frac{R_{p_o}}{4} = 0.5R_{p_o}$$

So possible inputs that cause worst-case rise resistance:

- 0011
- 0101
- 0110

2) Fall Resistance: worst-case fall resistance occurs when the output node discharges through NMOS B, C, D:

$$R|_{fall,worst} = \frac{R_{n_o}}{1} + \frac{R_{n_o}}{2} + \frac{R_{n_o}}{2} = 2R_{n_o}$$

Inputs = 0111